Assignment # 4

Artificial Intelligence

Total Marks: 80 **Submission Date:** 17-06-2026

Name **Roll No.**

Instructions:

I. Assignment should be hand written and in a group of three members.
II. You should create your own solution according to your logic. Any similarity with other group and internet will be consider as cheating and your assignment will be marked as ZERO.

Question#1:

Consider the following dataset for the Naïve Baves Classifier:

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

Now calculate the probability of the new instance that to which class Mammals or non-mammals it belongs. Show complete working.

Question#2:

You are given a binary classification problem with the following training dataset:

Sample	x1	x2	Bias ($x_0 = 1$)	Target
1	2	1	1	1
2	-1	-3	1	-1
3	1	-1	1	-1

Perform **Simple Perceptron** over the dataset and show the weight updates after each misclassified sample.

Suppose all initial weights are 0 and alpha is 1.

Compute final output using following function show the final weights.

$$\begin{cases} 1 & \text{if net} > 0 \\ -1 & \text{otherwise} \end{cases}$$

Question#3:

Consider the following training dataset for KNN algorithm:

Movie	Explosions	Laughter	Genre
M1	10	2	Action
M2	5	8	Comedy
M3	8	3	Action
M4	3	7	Comedy
M5	12	1	Action
M6	4	6	Comedy

Now, let's say we have a new movie, M7, with features (Explosions = 6, Laughter = 4), and we want to classify it into "Action" or "Comedy" using KNN with k = 3 (considering the three nearest neighbors) and k = 5 (considering the five nearest neighbors). Use the Euclidean Distance or Manhattan Distance to predict the output of new instance, M7. Show complete working.

Question#4:

Consider the following dataset for the customers:

Customer	Annual Income	Spending Score
C1	40,000	60
C2	60,000	40
C3	80,000	70
C4	20,000	5
C5	120,000	90
C6	100,000	60

Lets assume the two cluster centroids:

Centroid 1: (40,000, 60)

Centroid 2: (80,000, 70)

Using the Euclidean Distance Formula, determine the two final clusters after the two iterations of the K means algorithm. You should terminate the process after two iterations.

----------**End of Assignment** ----------